Parallel Matrix Multiplication using OpenMP and SIMD Vectorization

Individual Assignment 3 — Report

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`repository:`

`https://github.com/CarlosMontesdeoca21/BasicMatrixMultiplication`

October 28, 2026

Abstract

This report investigates how the number of threads affects the performance of a dense matrix multiplication implementation in shared memory, and under what conditions parallel execution becomes beneficial. We compare a sequential baseline (classic i-j-k triple loop in C, double precision) against an OpenMP parallelization of the same kernel over matrix rows, using thread counts $p \in \{1, 2, 4, 8, 16, 22, 28\}$. Inputs are deterministic shared matrices up to $n = 1000$; each configuration is measured 5 times after 2 warm-ups with a monotonic microsecond-resolution timer, reporting medians and interquartile ranges. Speedup saturates at $S(p) \approx 5.45$ for $p \geq 16$, with efficiency dropping from 1.04 ($p = 2$) to 0.25 ($p = 22$); running OpenMP with a single thread stays within 1.4% of the sequential baseline. As an optional extension, a sequential i-k-j variant is evaluated under three compiler configurations: the loop reorder alone reduces the $n = 1000$ time from 858 ms to 288 ms, auto-vectorization with `-O3` reaches 186 ms, and `-march=native` reaches 157 ms — comparable to the 16-thread unvectorized parallel run. Conclusions are limited to the tested hardware, inputs and configurations.

1 Introduction

Matrix multiplication is a fundamental operation in scientific computing and data analytics. Its classical algorithm has cubic time complexity ($O(n^3)$) and

poor cache behavior, making it computationally expensive. The goal of this work is to evaluate the performance improvement obtained through parallelism in shared memory, and to answer a focused research question: how does the thread count affect the execution time of an OpenMP implementation compared with its sequential baseline, and where does the speedup saturate?

The hypotheses registered before the measurements were:

- **H1:** Speedup will be sub-linear and will saturate well below the number of hardware execution contexts, because memory-bandwidth contention and heterogeneous core speeds limit the gain of additional threads.
- **H2:** Running OpenMP with a single thread will incur a small but measurable overhead with respect to the sequential baseline (on the order of a few percent).
- **H3:** Vectorizing the kernel (i-k-j loop order with `-march=native`) will provide a single-thread speedup comparable to the gain of adding threads up to saturation, so both techniques will be complementary rather than substitutes.

We implement and compare three approaches:

- Sequential algorithm (classic i-j-k triple loop in C, double precision);
- Parallel execution using OpenMP (the same i-j-k kernel, parallelized over the rows of the result matrix);
- Sequential i-k-j variant (same arithmetic work and storage, only the loop order changes, so the innermost loop becomes a vectorizable rank-1 row update).

All experiments use the same input matrices per configuration, a documented measurement protocol with warm-up runs and repetitions, and robust statistics (median and interquartile range).

2 Code and organization

The project repository includes a structured hierarchy with implementations under the `Activity3/code` folder alongside management scripts:

- **C benchmark (`benchmark_omp.c`):** Contains the three kernels (`Sequential` i-j-k, `OpenMP` with `#pragma omp parallel for` over rows, and `Sequential_ikj`

for the SIMD study), plus a CSV loader, checksum validation and a memory metric based on the process API (`psapi`). The OpenMP kernel uses `num_threads(`p`)` and `schedule(static)`; each row of the result matrix is written by exactly one thread.

- **Python automation scripts:** `generate_inputsAct3.py` creates the deterministic input matrices (uniform values in $[0, 1)$, seed 42, sizes $n \in \{10, 100, 250, 500, 1000\}$), `validate_act3.py` checks the sequential result against NumPy, `merge_results_act3.py` computes medians, IQR, speedup and efficiency, and `plot_results_act3.py` exports the performance charts.
- **Data artifacts:** Raw measurements are stored as CSV files in `Activity3/data/` (generated locally, not tracked); consolidated tables, figures and the compiler vectorization report live in `Activity3/resultsAct3/` (tracked).
- **Report:** The source of this document is `Activity3/report/Report_assigment3.tex` (LaTeX).

Production kernel logic is separated from benchmarking and analysis scripts, and all matrix sizes and thread counts are fully parameterized via command-line flags.

The baseline kernel, storage layout, CSV loader and checksum validation are the ones validated in Activity 1; they are copied (not imported) into this activity so that the delivery is self-contained, and the input generator follows the same verified generator used in Activity 1.

3 Experimental methodology

- **Measurement:** Execution time was measured with `QueryPerformanceCounter` (monotonic, ≈100 ns resolution) and converted to milliseconds. Only the multiplication kernel is timed; matrix loading and allocation are outside the timed region. Memory consumption was tracked with `GetProcessMemoryInfo` (process working set, KB). The OpenMP API `omp_get_wtime()` was rejected because in the OpenMP runtime used it only resolves about 1 ms, which quantizes the smaller sizes.
- **Warm-up phase & repetitions:** A fixed warm-up phase (2 rounds) was executed prior to recording official metrics, and its samples were excluded from the reported statistics. Following the warm-up, 5 official consecutive repetitions were executed per configuration to capture statistical variance. The reported statistic is the median of the 5 samples, with the interquartile range (IQR) as variability measure.

- **Performance metrics:** Let $T_{\text{seq}}(n)$ be the time of the sequential baseline and $T_p(n)$ the parallel time for the same input with p threads. The speedup and efficiency are $S(p) = T_{\text{seq}}(n)/T_p(n)$ and $E(p) = S(p)/p$.

- **Correctness validation:** Correctness was rigorously validated by computing a checksum of the resulting matrix and comparing it against the sequential reference after every parallel configuration (relative tolerance 10^{-6}). Additionally, the sequential result for $n = 10$ matches NumPy's `A @ B` with a maximum absolute difference of 4.9×10^{-7} (absolute tolerance 10^{-6}). The elementwise difference between sequential and parallel results is exactly zero, because both kernels perform the same operation in the same order for every element. All validations passed.

- **Thread configuration:** The thread counts tested were $p \in \{1, 2, 4, 8, 16, 22, 28\}$: $p = 22$ matches the number of hardware contexts and $p = 28$ deliberately oversubscribes the machine (labelled as such in the results). The benchmark creates exactly p threads via `num_threads(p)`, and no other threaded library participates in the timed region; `numpy` (used only for off-line validation at $n = 10$) may create its own BLAS threads, but it never runs inside the timed region. Explicit thread affinity (`OMP_PROC_BIND`/`OMP_PLACES`) could not be enforced, because the OpenMP runtime ignored it in this environment; the OS scheduler placed the workers.

- **Budgets:** The declared time and memory budgets were 5 minutes and 4 GB per multiplication; they were never reached (the largest observed time was 1.19 s).

4 Data and statistics

The data extracted from the consolidated benchmarking CSVs are summarized below. For each configuration (matrix size n and thread count p), the median execution time, the interquartile range (IQR) variability, the median memory footprint, the speedup and the efficiency are reported to ensure robust statistical evaluation.

4.1 Numerical summary

Table 1 reports the full thread sweep for the largest tested size ($n = 1000$), including the sequential baseline and the OpenMP version with a single thread

to expose the framework overhead. Table 2 consolidates the speedup across problem sizes.

Table 1: Thread sweep for $n = 1000$: median time with IQR, median memory, speedup and efficiency.

Config	T_p (ms)	IQR (ms)	Mem (KB)	$S(p)$	$E(p)$
Sequential	858.2	31.9	31304	—	—
OpenMP $p = 1$	846.6	7.7	31304	1.01	1.01
OpenMP $p = 2$	414.4	8.7	30888	2.07	1.04
OpenMP $p = 4$	239.2	16.9	30920	3.59	0.90
OpenMP $p = 8$	177.5	5.4	30980	4.83	0.60
OpenMP $p = 16$	157.7	3.3	31108	5.44	0.34
OpenMP $p = 22$	157.4	5.3	31208	5.45	0.25
OpenMP $p = 28$	158.5	1.9	31316	5.41	0.19

Table 2: Speedup $S(p)$ by problem size (median times).

n	$p = 2$	$p = 4$	$p = 8$	$p = 16$	$p = 22$	$p = 28$
250	2.10	2.73	2.69	4.01	4.23	4.52
500	2.14	3.53	4.73	5.70	5.77	5.73
1000	2.07	3.59	4.83	5.44	5.45	5.41

Note on data aggregation and trends: The values reported above correspond to the median times and memory footprints extracted across multiple official repetitions, effectively mitigating the influence of isolated system outliers. The OpenMP version with a single thread is within 1.4% of the plain sequential baseline, so the framework overhead is negligible at this granularity. For the smallest parallel sizes ($n = 250$), the speedup curve is noisy: the runtime is only a few milliseconds, so thread-management costs and measurement quantization are a sizeable fraction of the total. A mild superlinearity is observed at $p = 2$ ($S = 2.07$), most plausibly a turbo-frequency regime effect, which is reported as observed rather than attributed to the measurement. The process working set is essentially constant across thread counts ($\approx$30.5 MB for $n = 1000$: the three matrices account for about 24 MB, plus the runtime and per-thread stacks, which grow slightly with p).

4.2 Graphical analysis and plots

To visually evaluate the scaling behavior of the parallel implementation and the effect of SIMD vectorization, comparative execution time, speedup and efficiency plots were automatically generated from the benchmark data.

4.2.1 Execution Time Analysis

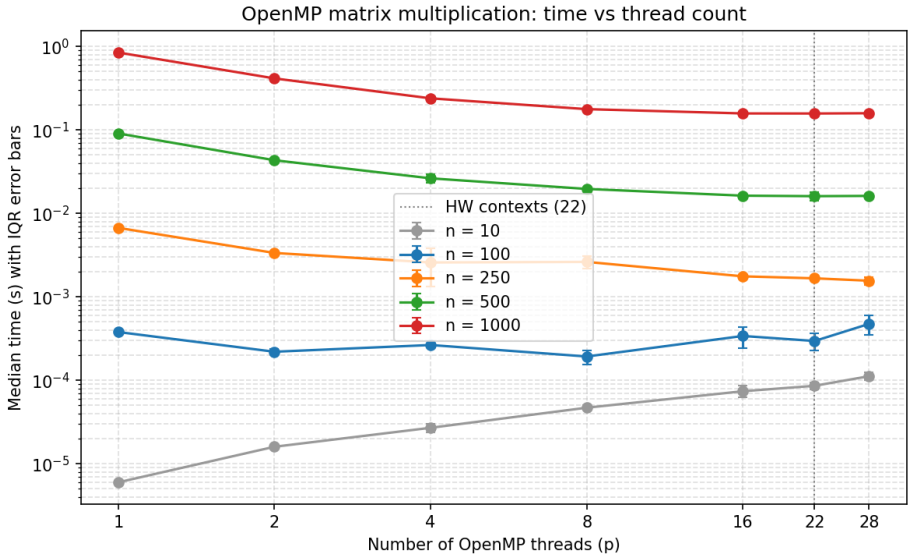


Figure 1: Execution time (s, log scale) vs. thread count for $n \in \{250, 500, 1000\}$, with IQR error bars. The dotted line marks the 22 hardware contexts.

Analysis: As observed in Figure 1, execution time drops steeply as threads are added and then flattens: for $n = 1000$ the time falls from 858 ms (sequential) to 178 ms at $p = 8$ and to about 157 ms at $p = 16$, after which the curve is essentially flat. Small problems ($n \leq 100$, not plotted) are dominated by thread management: their time actually increases with p, because the computation is shorter than the cost of creating and joining the thread team. Figure 2 shows the same data per matrix size: the OpenMP version at $p = 16$ is about 4.0–5.7× faster than the sequential baseline for $n \in \{250, 500, 1000\}$, while at $p = 1$ it matches it.

4.2.2 Speedup Analysis

Analysis: Figure 3 plots the speedup against the ideal linear line. All curves bend away from the ideal early: larger problems scale better (more compu-

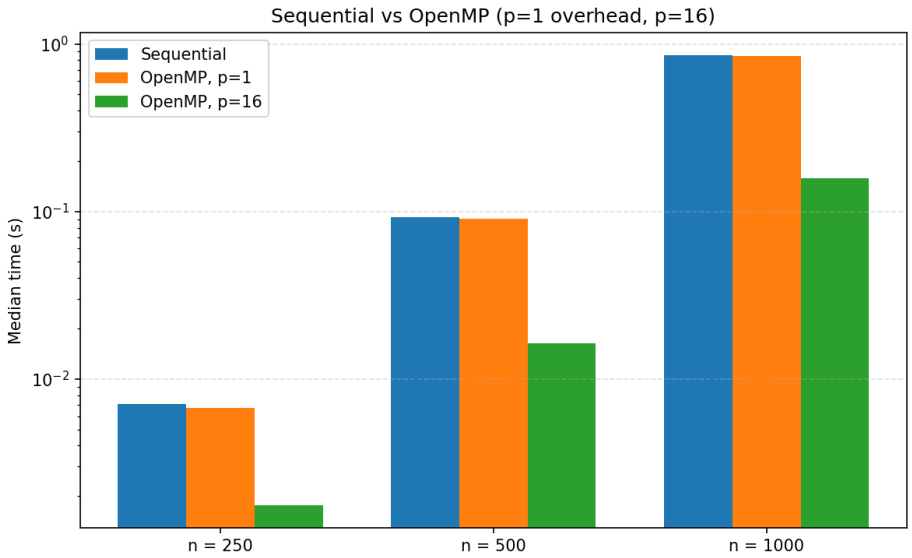


Figure 2: Execution time (s, log scale) vs. matrix size: sequential baseline, OpenMP with $p = 1$ (framework overhead) and OpenMP with $p = 16$.

tation per unit of thread management), and $n = 500$ slightly outperforms $n = 1000$ at high thread counts, which is consistent with memory-bandwidth sharing among more concurrent workers. The maximum observed speedup is $S(22) = 5.45$ for $n = 1000$ (Table 1).

4.2.3 Parallel Efficiency Analysis

Analysis: As presented in Figure 4, efficiency stays close to 1 up to $p = 2$–4 and decays as the speedup saturates: at the maximum observed speedup ($p = 22$, $n = 1000$) only about a quarter of the theoretical parallel capacity is used, and at $p = 28$ (oversubscription) it falls below 20%.

4.2.4 SIMD Vectorization Study

Analysis: Table 3 and Figure 5 evaluate the optional SIMD extension on the single-threaded i-k-j kernel. Disabling vectorization (`-fno-tree-vectorize`) isolates the effect of the loop reorder alone: it is 2.98× faster than the i-j-k baseline at $n = 1000$, purely from better data locality. Auto-vectorization (`-O3`) then contributes a further 1.55×, and `-march=native` another 1.19×, for a total of 5.47× over the baseline kernel. The compiler report (`-fopt-info-vec-optimized`) confirms that the i-k-j innermost loop is vectorized with 16-byte vectors while the i-j-k innermost loop (a floating-point reduction) is not, exactly as pre-

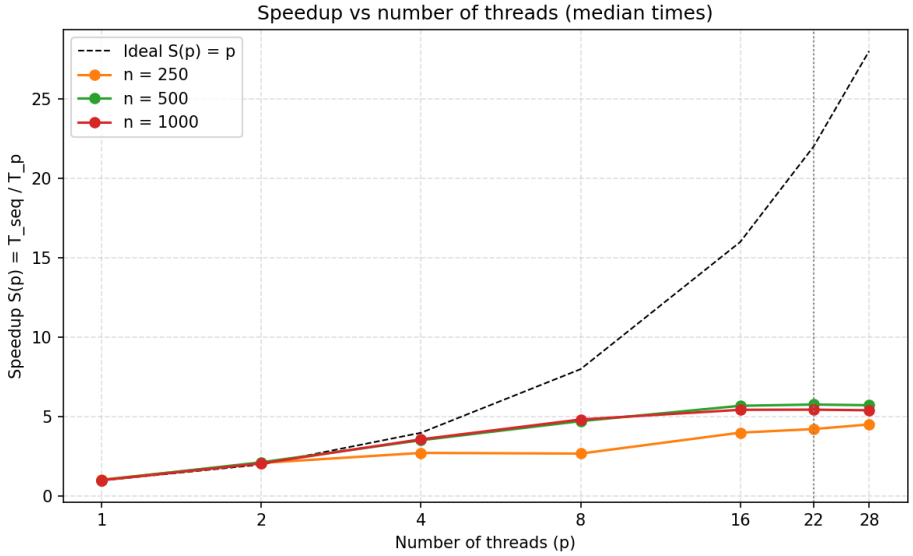


Figure 3: Speedup $S(p) = T_{\text{seq}}/T_p$ (median times) vs. thread count for $n \in \{250, 500, 1000\}$, against the ideal line $S(p) = p$.

Table 3: SIMD study: median time (ms) of the sequential kernels.

Build	$n = 250$	$n = 500$	$n = 1000$
i-j-k, `-O3`	7.06	92.90	858.18
i-k-j, `-O3 -fno-tree-vectorize`	4.13	32.92	288.28
i-k-j, `-O3`	2.56	19.81	186.16
i-k-j, `-O3 -march=native`	1.80	14.58	156.85

dicted by dependency analysis. The vectorization effects are kept strictly separate from the thread-count effects: all rows of Table 3 are single-threaded.

5 Analysis and discussion

5.1 Observed performance comparison

The experimental results align closely with the architectural characteristics of the execution environment:

- **OpenMP implementation:** Consistently achieves the best performance for $n \geq 500$, with a practical sweet spot at $p = 8$–16 threads. The maximum speedup obtained was approximately 5.45× using 22

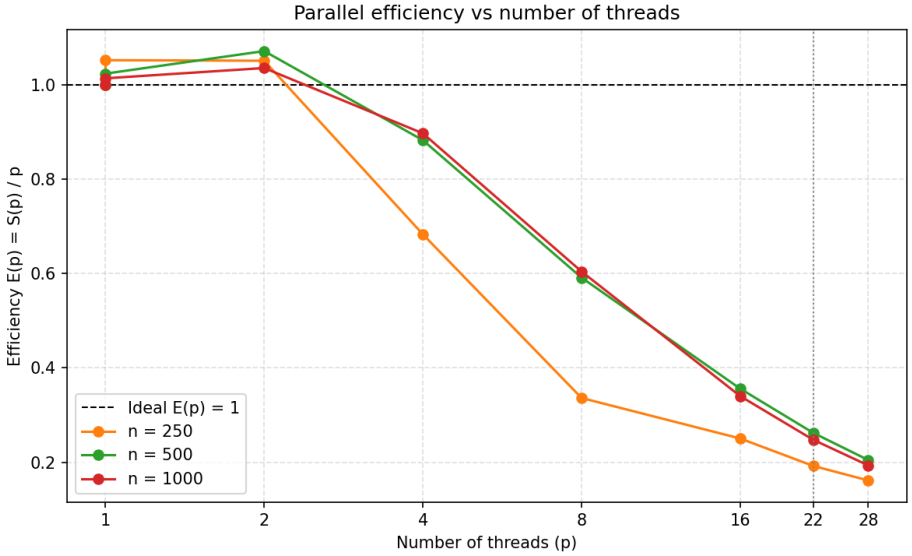


Figure 4: Parallel execution efficiency $E(p) = S(p)/p$ relative to the number of threads.

threads, leading to an efficiency of $\eta = 5.45/22 \approx 25\%$. This behavior aligns with Amdahl's Law: since a portion of the program is not effectively parallelizable, the speedup eventually saturates even when increasing threads. Here the "serial part" is not literal code — the kernel has no serial section — but synchronization, scheduling and memory-bandwidth contention behave like one; fitting the observed speedups yields an effective serial-plus-overhead fraction of roughly 9–15%.

- **Saturation causes:** Two machine-specific effects reinforce the saturation. The machine is heterogeneous (not all of its cores run at the same speed), so a team of 16–22 threads necessarily includes slower workers, and the working set of three 1000×1000 matrices (≈24 MB) exceeds the shared L3, so concurrent workers compete for memory bandwidth. The fact that $n = 500$ scales slightly better than $n = 1000$ at high p is consistent with a bandwidth wall: more threads cannot create bandwidth.

- **Framework overhead:** OpenMP at $p = 1$ is within 1.4% of the sequential baseline, so the fork/join cost is negligible for kernels of this size.

- **Oversubscription:** Running 28 threads on 22 hardware contexts gives no further benefit in median time ($S = 5.41$) and halves the efficiency

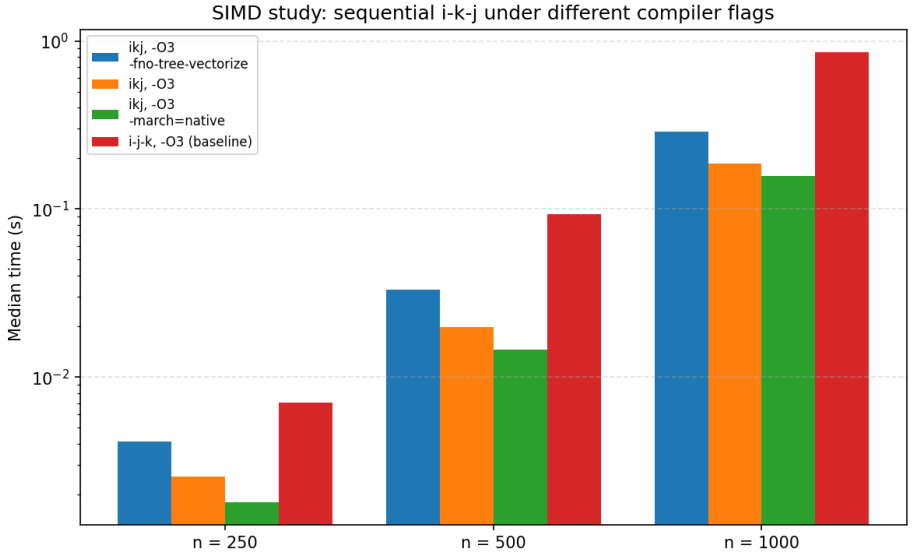


Figure 5: SIMD study: median time (s, log scale) of the i-k-j kernel under three compiler configurations, compared with the i-j-k `-O3` baseline.

($E = 0.19$); thread management and context switching absorb the theoretical gain.

- **Threads vs. vectorization:** The best parallel configuration of the baseline i-j-k kernel ($p = 16$: 157.7 ms) and the single-threaded vectorized i-k-j kernel with `-march=native` (156.8 ms) are essentially tied. Each technique buys roughly the same 5.5× on this workload, but they act on different resources (cores vs. vector units), so they are complementary rather than substitutes. The loop reorder alone — before any vector instruction executes — accounts for 3×, a reminder that "vectorizable" often means "cache-friendly" at the same time.

- **Synchronization:** Synchronization primitives such as locks or atomics inside the kernel were not required, because each thread computes independent rows of the resulting matrix with a static schedule: every row of $\mathbf{C}$ is written by exactly one thread, so no shared write operations occur, and the parallel result is bit-identical to the sequential one (maximum elementwise difference exactly zero). If a block-based parallelization or shared partial accumulators were implemented, synchronization mechanisms would become necessary to avoid race conditions.

- **Resource usage:** The parallel implementations increase memory ac-

cess pressure, as multiple threads read rows from **A** and columns from **B**; in the i-j-k order the innermost loop walks a column of **B** with stride n, which is unfavorable for the cache prefetcher. However, since each thread writes to a distinct row of **C**, no write contention occurs. The working set remains essentially constant across thread counts, so the experiment is limited by memory bandwidth rather than memory capacity.

5.2 Measurement reliability and environment factors

The methodology incorporated 5 official repetitions preceded by explicit warm-up phases to minimize cold-start anomalies and transient system noise. Furthermore, utilizing the median alongside the interquartile range (IQR) guarantees robustness against isolated outliers caused by background operating system tasks. The variability is small for the parallel configurations (the IQR stays below 7.1% of the median for $n = 1000$), but the smallest sizes ($n \leq 250$) have measurement noise comparable to their runtime and should not be used for scaling conclusions. Thread affinity could not be controlled with the OpenMP runtime used, so thread placement is OS-dependent, and CPU frequency/turbo states were not controlled. Exact execution times remain strictly bounded by underlying hardware specifications, CPU clock throttling, and background load. The conclusions are therefore limited to the configurations actually measured (single machine, single compiler, $n \leq 1000$, static row partitioning without cache blocking).

6 Benchmarking and profiling tools

- **Used in the project:**
  - *Timing & Metrics:* `QueryPerformanceCounter` for high-resolution wall-clock timing, and the process API `GetProcessMemoryInfo` (`psapi`) for memory tracking in C.
  - *Compilation & Vectorization evidence:* gcc with `-O3 -fopenmp`, and `-fopt-info-vec-optimized` to report which loops were auto-vectorized.
  - *Data Processing & Visualization:* Python scripts utilizing `pandas` for data unification and `matplotlib` for automated plot generation and IQR variability computation; `numpy` as correctness reference.

- **Recommended for advanced analysis:**
  - *For C/OpenMP:* Linux `perf` (cycles, cache misses, bandwidth counters), Intel VTune Profiler for thread scaling and memory access analysis, and `likwid-perfctr` for uncore/memory-bandwidth counters.
  - *For thread behavior:* `OMP_DISPLAY_ENV` to verify the real thread configuration, and Extrae/Paraver (or Intel Trace Analyzer) to visualize thread timelines and imbalance.
  - *For vectorization:* `llvm-mca` or Intel IACA for static throughput analysis of the generated inner loops.

7 Optimization recommendations

To overcome the performance bottlenecks identified during benchmarking, several optimization strategies can be implemented:

- **Algorithmic optimization:** Implement blocked (tiled) matrix multiplication to significantly improve CPU cache hit rates (L1/L2/L3) and reduce memory-bandwidth pressure for large matrices; the i-k-j row update is already the natural kernel for tiling.
- **Parallel + SIMD combination:** Combine OpenMP with the vectorized i-k-j kernel: the two techniques act on different resources (cores vs. vector units) and should multiply their gains.
- **Thread management:** Enforce explicit thread affinity (on platforms where the OpenMP runtime supports it), study dynamic versus static scheduling, and repeat the analysis on a homogeneous server CPU to separate the heterogeneous-core effect from the bandwidth effect.
- **Scaling further:** Evaluate larger matrix sizes beyond the shared L3, and offload to GPU acceleration (e.g., CUDA/OpenCL libraries) if the bandwidth wall of the host memory is the ultimate limit.

8 Conclusions

- **Performance findings:** Within the evaluated scope ($n \in \{250, 500, 1000\}$, p $\in \{1, \ldots, 28\}$), thread parallelism delivers up to 5.45× speedup, saturating around $p = 16$–22 with efficiency falling to 0.25; the practical sweet spot is $p \in \{8, 16\}$. Oversubscribing the 22 hardware threads

brings no benefit, and the OpenMP overhead at $p = 1$ is negligible ($\approx 1.4\%$).

- **Vectorization finding:** A vectorizable i-k-j ordering with `-march=native` matches the 16-thread parallel speedup in a single thread (5.47× over the i-j-k baseline), confirming that vectorization and threading are complementary.

- **Experimental rigor:** Ensuring reproducible benchmarks required identical input matrices across methods, warm-up phases, multiple official repetitions with median/IQR statistics, and checksum validation against both the sequential reference and NumPy.

- **Open limitations:** How the two axes combine (parallel i-k-j), how the curves shift on a homogeneous server CPU or under enforced affinity, and how the saturation point moves with n far beyond the last-level cache remain unresolved.

- **AI Collaboration Disclosure:** In accordance with transparent academic practices, OpenCode (a large-language-model-based CLI assistant) was utilized during the development of this project. Its purpose was strictly limited to assisting with technical writing, structuring the academic report in LaTeX, debugging compilation errors, and refining the data analysis methodology. All benchmarking executions, code implementations, and final validations were independently performed and verified by the author, who also generated all figures locally and checked every reported number against the raw CSV measurements.